Week 11

Hierarchical, Asynchronous, and Decentralized Federated Learning

Update routes, timing, and client influence

Graduate course · English instruction



Faranak Solat
Department of AI & Data Science
Sejong University

Why Change the Architecture?

12 phones

Each phone can train one local m odel.

3 hospitals

Nearby devices share a gateway.

1 cloud link per site

Wide-area uploads and slow devic es can both delay training.

Which cost comes from the route, and which comes from waiting?

Ludwig & Baracaldo (2022), Chs. 9–12 · Kairouz et al. (2021), §§2.1 & 7

Learning Objectives

Trace the route

Mark where each update is combi ned.

Read the clock

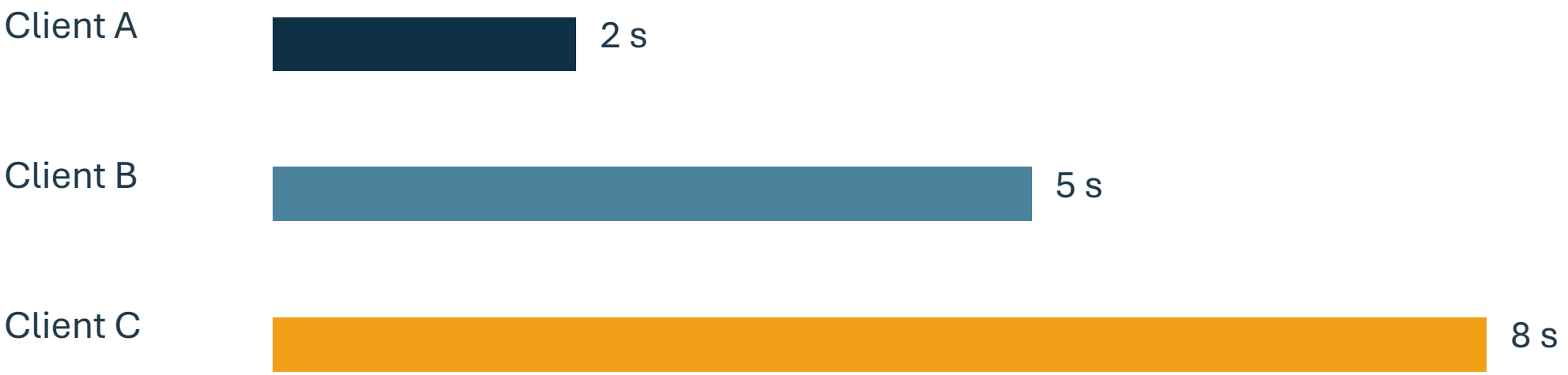
Identify the waiting rule and calcul ate the age of a late update.

Check the weights

Predict which clients have more in fluence on the model.

Ludwig & Baracaldo (2022), Chs. 9–12 · Kairouz et al. (2021), §§2.1 & 7

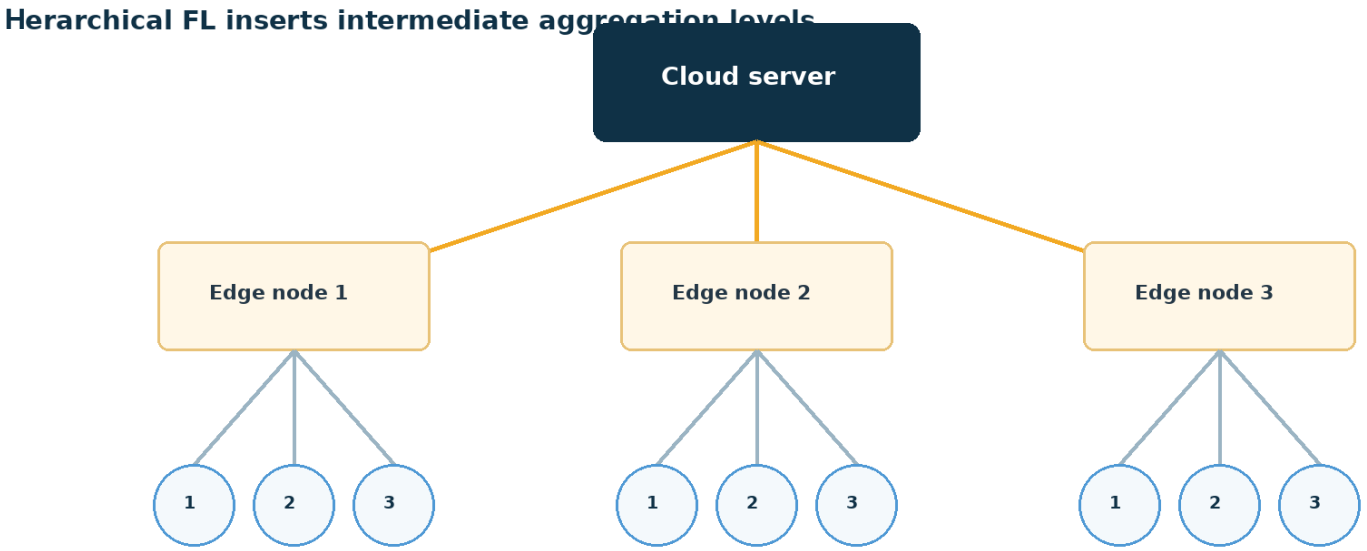
Synchronous Client–Server FL



Wait for all three: 8 seconds. Accept the first two: 5 seconds, with C missing.

Ludwig & Baracaldo (2022), Chs. 9–12 · Kairouz et al. (2021), §§2.1 & 7

Hierarchical Federated Learning



Sample-weighted average

Gateway A: 20 examples, value 2
Gateway B: 80 examples, value 5

$(20 \times 2 + 80 \times 5) / 100 = 4.4$

Equal gateway weights would give 3.5. The sample counts change the result.

Ludwig & Baracaldo (2022), Chs. 9–12 · Kairouz et al. (2021), §§2.1 & 7

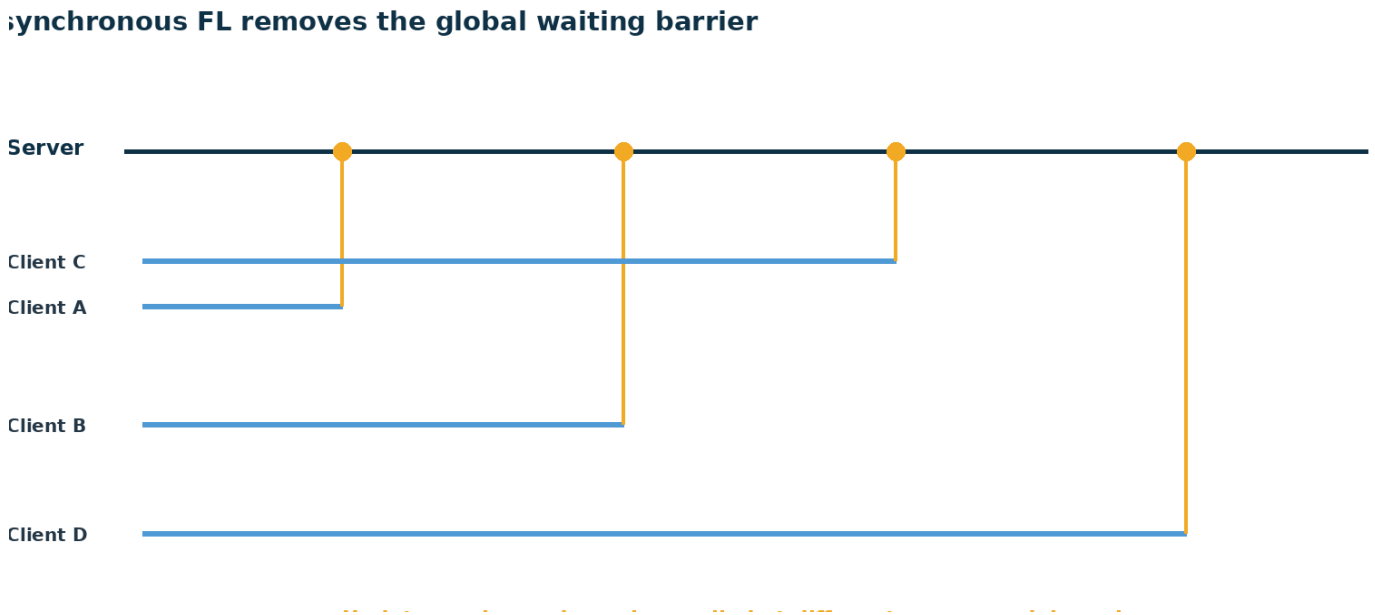
Why Hierarchy Can Help

Route	Local upload	Cloud upload
12 phones → cloud	0 MB	12 × 1 MB = 12 MB
12 phones → 3 gateways → cloud	12 × 1 MB = 12 MB	3 × 1 MB = 3 MB

Cloud uploads fall by 75%. Total uploads rise from 12 MB to 15 MB.

Ludwig & Baracaldo (2022), Chs. 9–12 · Kairouz et al. (2021), §§2.1 & 7

Asynchronous Federated Learning



Follow the late client

A finishes first and changes the serve r model. B returns later with a model trained on the earlier version.

Record the starting version of each client model.

Ludwig & Baracaldo (2022), Chs. 9–12 · Kairouz et al. (2021), §§2.1 & 7

Staleness: An Update Can Be Computed on an Older Model

Version gap

Current version 12
Starting version 9
Gap = 12 − 9 = 3

Example mixing rule

Weight = 0.4 / (1 + gap)
At gap 3, weight = 0.1

Server update

Current value 10
Client value 6
0.9 × 10 + 0.1 × 6 = 9.6

If slow clients repeatedly receive small weights, check their test accuracy.

Ludwig & Baracaldo (2022), Chs. 9–12 · Kairouz et al. (2021), §§2.1 & 7

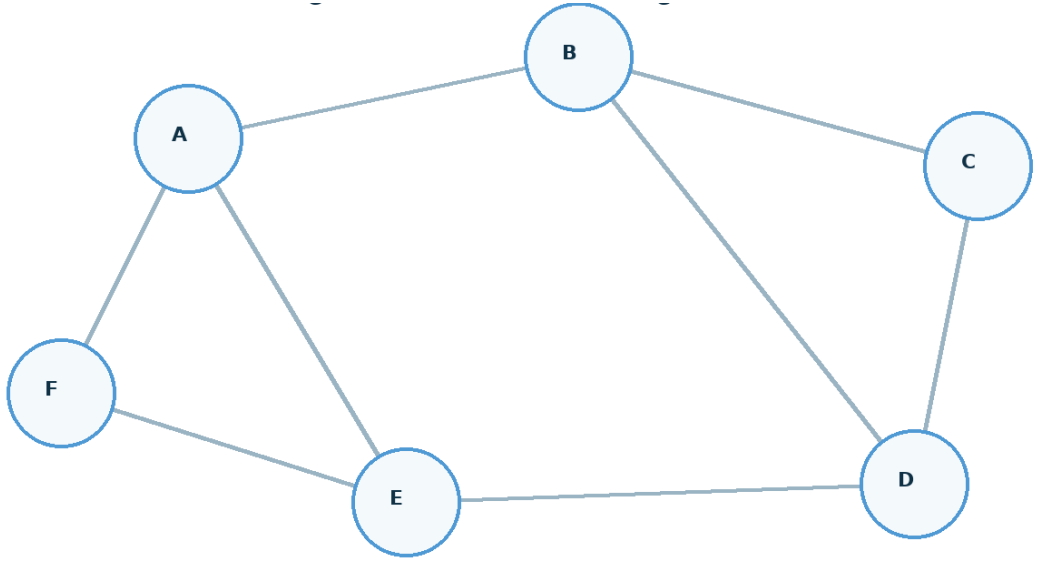
Synchronous vs. Asynchronous FL

Question	Synchronous	Asynchronous
When does the model change?	After the required replies	On each accepted arrival
Which delay matters?	Time to close the round	Age of each arriving model
What should we record?	Missing and late clients	Version gap and update count per client

Compare accuracy against elapsed time and against training work.

Ludwig & Baracaldo (2022), Chs. 9–12 · Kairouz et al. (2021), §§2.1 & 7

Decentralized / Peer-to-Peer Federated Learning



One peer exchange

A holds 2. B holds 6. With equal mixing weights, both get (2 + 6) / 2 = 4. Other peers still hold their earlier values.

Information spreads one active link at a time.

Ludwig & Baracaldo (2022), Chs. 9–12 · Kairouz et al. (2021), §§2.1 & 7

Architecture Trade-offs

Design	Communication route	Main design choice
Synchronous server	Client ↔ server	How many replies close a round?
Hierarchical	Client ↔ gateway ↔ cloud	How often should gateways synchronize?
Asynchronous server	Client ↔ server	How much weight should late updates get?
Decentralized	Peer ↔ neighbors	Which links and mixing weights are used?

A hierarchy can also use asynchronous updates at the cloud.

Ludwig & Baracaldo (2022), Chs. 9–12 · Kairouz et al. (2021), §§2.1 & 7

How to Choose an Architecture

Busy cloud link

Measure cloud-facing upload byte s for direct and gateway routes.

Unpredictable finish times

Measure time to target accuracy a nd participation by client.

No central coordinator

Check connectivity and agreemen t among neighboring models.

WEEK 11

In-Class System Design

Scenario

Three hospitals train with local wearables. One hos pital loses its cloud connection for two rounds. Its phones can still reach their gateway.

Two-minute design task

Draw the message route. Choose a local waiting rule. Deci de what version and sample count the gateway should sen d when it reconnects.

Defend the design with one quality metric and one systems metric.

Ludwig & Baracaldo (2022), Chs. 9–12 · Kairouz et al. (2021), §§2.1 & 7

Key Takeaways

9 MB

Cloud upload saved per step in th e twelve-phone example.

3 versions

Age of the update trained on versi on 9 and received at version 12.

4

New value at both peers after equ al mixing of 2 and 6.

Reading & Preparation

Course reading

Ludwig & Baracaldo (eds.), Federated Learning (2022), chapters 9–12. Kairouz et al., Advances and Open Problems in Federated Learning (2021), sections 2.1 and 7.

Preparation

Trace one update. Name its sender, starting model version and aggregation weight.

Q & A

Thank You.

sofar@sejong.ac.kr